



## S1.5 Ideal gases (HL)

### Structure 1 · Models of the particulate nature of matter · IB Diploma · Concept guide and original exam-style practice

#### Understand the concept

An ideal gas is modelled as point particles with negligible volume, no intermolecular forces, and in constant random motion undergoing perfectly elastic collisions; real gases approximate this behaviour best at low pressure and high temperature. The ideal gas law,  $pV = nRT$  ( $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$ , with  $p$  in Pa,  $V$  in  $\text{m}^3$ ,  $T$  in K), relates pressure, volume, amount and temperature, and combines Boyle's, Charles's and Avogadro's laws. At standard temperature and pressure, one mole of any ideal gas occupies the same molar volume, allowing gas volumes to be used directly in stoichiometric calculations.

#### Worked example: follow the reasoning

Calculate the volume, in  $\text{dm}^3$ , occupied by 2.40 g of hydrogen gas,  $\text{H}_2$  ( $M = 2.02 \text{ g mol}^{-1}$ ), at 25 °C and 100 kPa.

Step 1:  $n(\text{H}_2) = 2.40 / 2.02 = 1.188 \text{ mol}$ .

Step 2: Convert units:  $T = 25 + 273 = 298 \text{ K}$ ;  $p = 100\,000 \text{ Pa}$ .

Step 3: Rearrange  $pV = nRT \rightarrow V = nRT/p = (1.188 \times 8.31 \times 298) / 100\,000$ .

Step 4:  $V = 2942 / 100\,000 = 0.0294 \text{ m}^3$ .

Step 5: Convert to  $\text{dm}^3$ :  $0.0294 \text{ m}^3 \times 1000 = 29.4 \text{ dm}^3$ .

Answer:  $V \approx 29.4 \text{ dm}^3$ .

#### Quick check · multiple choice

A fixed mass of ideal gas has its absolute temperature doubled while the volume is halved. What happens to the pressure?

- A It is halved
- B It stays the same
- C It doubles
- D It quadruples

Write your choice: \_\_\_\_\_

### Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A rigid  $5.00 \text{ dm}^3$  cylinder contains an ideal gas at  $20.0 \text{ °C}$  and a pressure of 202 kPa. [6]
- (a) Calculate the amount, in mol, of gas in the cylinder. [2]
  - (b) The cylinder is heated to  $85.0 \text{ °C}$  without any gas escaping. Calculate the new pressure in the cylinder. [2]
  - (c) State two assumptions of the ideal gas model that may not hold for a real gas at high pressure. [2]
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Final answer: \_\_\_\_\_



## Answers and explanation

### Question 1 [6]

(a)  $T = 293 \text{ K}$ ,  $p = 202\,000 \text{ Pa}$ ,  $V = 5.00 \times 10^{-3} \text{ m}^3$ .  $n = pV/RT = (202\,000 \times 5.00 \times 10^{-3}) / (8.31 \times 293) = 1010/2435 = 0.415 \text{ mol}$ . [2]

(b) At constant  $n$  and  $V$ ,  $p/T = \text{constant}$ , so  $p_2 = p_1 \times (T_2/T_1) = 202 \times (358/293) = 202 \times 1.222 = 247 \text{ kPa}$ . [2]

(c) Any two of: real gas particles have finite volume (not negligible) which becomes significant at high pressure; there are intermolecular attractive forces between real gas particles (assumed zero in the ideal model), which are more significant when particles are forced close together at high pressure. [2]

### Quick check answer

D. From  $pV = nRT$ ,  $p = nRT/V$ ; doubling  $T$  and halving  $V$  both increase  $p$  by a factor of 2, so overall  $p$  increases by  $2 \times 2 = 4$ .

#### Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Chemistry HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.